The Rock Cycle and the Three Rock Types

Hi classmates! This is my reviewer for Earth and Life Science. The big idea is simple: rocks are not permanent. Over thousands to millions of years, any rock can be broken down, buried, squeezed, melted and rebuilt into a different kind of rock. That never-ending recycling of Earth materials is what we call the rock cycle.

What is a rock?

A rock is a naturally occurring solid made of one or more minerals, or sometimes of natural glass or bits of organic matter. A mineral, on the other hand, is a naturally formed, inorganic solid with a definite chemical makeup and an orderly internal arrangement of atoms. Think of minerals as the ingredients and rocks as the finished dish.

Geologists sort rocks into three families based on how they formed, not on their color or size.

Igneous rocks: born from fire

Igneous rocks form when molten rock cools and hardens. This hardening into mineral crystals is called crystallization. Molten rock underground is called magma; once it reaches the surface, it is called lava.

- Intrusive (plutonic) igneous rocks cool slowly deep underground. Slow cooling gives crystals time to grow, so the texture is coarse-grained. Granite and diorite are examples.
- Extrusive (volcanic) igneous rocks cool quickly at or near the surface. Fast cooling makes tiny crystals, so the texture is fine-grained. Basalt and andesite are examples.
- If lava cools almost instantly, there is no time for crystals at all, so you get volcanic glass like obsidian.
- Pumice is a light, bubbly extrusive rock full of holes left by escaping gas. It can even float on water.

The Philippines sits on the Pacific Ring of Fire, so igneous rocks are everywhere here. Andesite is very common in our stratovolcanoes like Mayon, and basalt shows up in many old lava flows. After the 1991 Pinatubo eruption, huge amounts of pumice and ash covered parts of Central Luzon.

Sedimentary rocks: layers of the past

Sedimentary rocks form from sediments, which are loose pieces of older rock, shells, or minerals that settled out of water. They usually form at or near the surface, and they are the rocks most likely to contain fossils.

The steps that make a sedimentary rock go like this:

- Weathering is the breaking down of rock in place. Mechanical weathering splits rock physically (for example, roots growing in cracks), while chemical weathering changes the minerals (for example, slightly acidic rainwater dissolving limestone).
- Erosion is the removal and transport of those broken pieces by water, wind, ice, or gravity.
- Deposition happens when the moving agent loses energy and drops the sediments, often in layers at the bottom of rivers, lakes or seas.
- Lithification turns loose sediment into solid rock. It has two parts: compaction (the weight of upper layers squeezes out water and space) and cementation (dissolved minerals glue the grains together).

Common examples are sandstone (cemented sand), shale (compacted mud or clay), conglomerate (rounded pebbles glued together) and limestone (mostly calcium carbonate, often from shells and coral). Coal is also sedimentary, formed from buried plant matter. In our country, the Chocolate Hills of Bohol and the dramatic cliffs of El Nido in Palawan are made of limestone, and the Puerto Princesa Underground River flows through a limestone cave system. Semirara Island in Antique is known for its coal mining.

Metamorphic rocks: changed, but not melted

Metamorphic rocks form when an existing rock (the parent rock) is transformed by high heat, high pressure, or chemically active fluids, while staying solid. If it melts, it is no longer metamorphism; that becomes the igneous path.

- Foliated metamorphic rocks have minerals lined up in bands or layers due to directed pressure. Examples: slate, schist and gneiss.
- Non-foliated metamorphic rocks have no banding. Examples: marble and quartzite.

Some parent-to-metamorphic pairs worth memorizing:

- Limestone becomes marble.
- Shale becomes slate.
- Sandstone becomes quartzite.
- Granite can become gneiss.

Romblon is famous in the Philippines for its marble, which is carved into tiles, tabletops and sculptures.

Putting the cycle together

The rock cycle has no fixed starting point, and rocks can take shortcuts. Here is one possible loop:

- Magma cools and crystallizes into igneous rock.
- Uplift exposes the rock; weathering and erosion break it apart.
- Sediments are deposited and lithified into sedimentary rock.
- Burial brings heat and pressure, producing metamorphic rock.
- Even more heat causes melting, forming magma again.

Shortcuts happen too. An igneous rock can be metamorphosed directly, and a metamorphic rock can be weathered into sediment without ever melting. The energy that drives the cycle comes from two sources: the Sun (powering weather, water movement and erosion at the surface) and Earth's internal heat (powering melting, plate movement and metamorphism).

Quick check

- Q: What is the term for molten rock found below Earth's surface? A: Magma.
- Q: Why does granite have larger crystals than basalt? A: Granite cools slowly underground, giving crystals more time to grow.
- Q: What two processes make up lithification? A: Compaction and cementation.
- Q: What metamorphic rock forms from limestone, and which Philippine province is known for it? A: Marble; Romblon.
- Q: What landform in Bohol is made of limestone? A: The Chocolate Hills.